KamiLimu x nLine: Data Science Case Studies

What students had learnt by August 2025

- Data manupulation using SQL queries
- Introduction to Python programming
- Python Libraries I: Data manipulation using Pandas, Numpy, Polars, Pyspark, GeoPandas
- Python Libraries II: Introduction to visualization, Data visualization best practices, Data visualization for professional settings, Python libraries for data visualization(Matplotlib, Seaborn, Plotly, Folium, GoePandas)
- Descriptive statistics: measures of central tendency, measures of spread, measures of distribution shape
- Inferential statistics: estimation using confidence intervals, hypothesis testing
- Probability and Statistics
- Time series analysis

- Machine Learning I: Types of ML, Supervised ML, Evaluating ML models

Objectives of the Case Study Exercise

- Provide students with real-world case scenarios with relevance and impact, where they can apply the skills they have learnt in the Data Science track so far.
- Deliver value to nLine by having students explore selected analysis problems through fresh perspectives, uncovering new insights and alternative approaches that can inform and enhance ongoing work.

Case Study 1: Prototyping a Result-Based Financing (RBF) Framework for Minigrid Investors

Context

Mini-grids are a critical solution for rural electrification, especially in Sub-Saharan Africa. In recent years, public and private sector actors have mobilized financing to support their deployment. However, investment decisions in the mini-grid sector are often made with limited access to reliable performance data. Moreover, there is a heavy reliance on metrics that may not accurately reflect the true reliability or quality of service experienced by end-users. As a result, funding mechanisms and policy frameworks risk being misaligned with actual mini-grid performance, potentially undermining efforts to improve energy access and long-term sustainability.

Result-Based Financing (RBF) offers a way to tie financial disbursement to verifiable performance outcomes. Various RBF mechanisms have been proposed and implemented—linking payments to KPIs such as system uptime, amount of energy delivered, and number of customer connections. However, these metrics often prioritize electrical connections over power quality. For example, rewarding

the number of customers connected to the grid doesn't account for whether the power delivered is stable, usable, or consistently available. A mini-grid that connects many households but delivers frequent undervoltage or long power outages may still receive funding—despite poor end-user experience.

Few RBF schemes incorporate high-resolution operational data or power quality indicators, and there's little consensus on which KPIs best reflect real-world performance. This presents an opportunity to revisit the design of RBF mechanisms using more granular data and meaningful KPIs.

Your Challenge

> ***Research existing RBF mechanisms in the energy access and/or mini-grid space, then prototype your own RBF mechanism that improves on the existing ones. Using the RBF mechanism you have developed together with the provided dataset (minigrid SAIDI, SAIFI, undervoltage hours, and number of connections), determine the payouts allocated to each minigrid.***

Data Provided

1 year of daily performance data from several mini-grids in North Kivu, DRC:

- SAIDI (average outage duration per customer)
- SAIFI (average outage frequency per customer)
- Hours of undervoltage

Guiding Questions

1. Identify some RBF mechanisms that exist for minigrid funding. Which organizations have used these mechanisms? Draw examples from minigrid development in African countries.
2. Identify which KPIs the minigrid investors/funders have used to reward or penalize minigrid developers.
3. What sources of data are used to determine these KPIs for minigrid developers? Are there any gaps/challenges in obtaining such datasets?
4. Are there any limitations to using such KPIs in evaluating minigrid performance? Are these KPIs sufficient to determine the impact of the

minigrids on customers' lived grid experiences?

5. Using the information from Q1-Q4
    a. Determine the KPIs that are best suited for rewarding/penalizing mini-grids.
    b. Develop an improved RBF mechanism that addresses the issues identified above and fairly rewards/penalizes minigrids based on performance and impact on customers' grid experiences.
6. Now use your developed RBF mechanism to reward/penalize the minigrids based on their performance (based on the provided minigrid dataset) and:
    a. Use your data manpulation and descriptive statistics skills to report on minigrid performance.
    b. Use visualizations to compare across sites or time.
    c. Try formulating a few "if-then" RBF payout conditions — e.g., *"payout if average SAIDI is below X for 3 months."*
7. Discuss why your new RBF mechanism is an improvement from the existing ones.

Deliverables

- A detailed report responding to the guiding questions.
- A data analysis notebook analyzing minigrid performance and implementing the RBF mechanism you have developed to determine payouts to different minigrids.
- A dashboard presenting the RBF mechanism.
- A short slide deck to present your work to potential investors.

Case Study 2: Operationalizing Grid Data for Cost Optimization in Utilities

Context

nLine collects granular, high-resolution grid performance data from distribution networks, offering daily insights into power quality and reliability at the secondary transformer level. Metrics like SAIDI (average outage duration), SAIFI (average outage frequency), and undervoltage hours are tracked consistently across distribution transformers.

This kind of data already supports the generation of summary reports and dashboards, which are useful for tracking long-term trends and informing periodic planning. But this data holds more opportunity: to use this data to support day-to-day operational decision-making for electricity distribution companies such as Ghana's ECG, Senegal's SENELEC, or Kenya's Kenya Power.

This data has the potential to power:

- Daily alerts for major grid events, such as large outages or severe voltage drops.
- Real-time prioritization by utilities of transformers for repair crew dispatch and scheduling.
- Investment planning in grid infrastructure based on chronic historical performance issues.
- Track the impact of grid interventions (upgrades, repairs, and maintenance) over time on grid performance.

Your Challenge

> ***This case study challenges you to take on the role of an electricity distribution comapny's data analytics team developing an actionable data-driven tool that helps operations teams use performance data to guide operations—from power outage response and monitoring, to long-term grid network planning.***

Data Provided

1 year of daily performance data from several transformers in Accra, Ghana:

- SAIDI (average outage duration per customer)

- SAIFI (average outage frequency per customer)
- Hours of undervoltage

Guiding Questions

- What are the roles of these teams in power distribution utilities such as Kenya Power (Kenya), ECG (Ghana), SENELEC (Senegal), and UMEME (Uganda). What specific functions do each of these teams perform?
    - system operations team
    - field services/ maintenance team
    - planning & engineering team
- Analyze the performance of the transformers based on SAIDI, SAIFI, and undervoltage hours.
- Identify which transformers are performing poorly on a daily basis and determine a repair and maintenance schedule for the field services team.
- Propose a scoring system for utility planners to use when allocating maintenance funds.
- Determine which transformers consistently underperform, and how this information should influence budget allocation for grid infrastructure upgrades.
- Formulate rules for grid upgrades and investment decisions from the power quality and reliability metrics?
- Identify areas/transformers with exemplary performance for operations bench-marking?
- How can a utility use your tool to reduce operational costs?

Utilize appropriate data analytics and visualization tools such as plots and maps to respond to these prompts.

Deliverables

- A detailed report responding to the guiding questions
- Exploratory data analysis notebook (with clear visualizations)

- A decision-support dashboard (scored/ranked transformer recommendations)
- A short slide deck to present your tool to an electricity distribution company e.g. Electricity Company of Ghana, Kenya Power, etc.

Case Study 3: Detecting Generator Usage from Sensor Data

Context

In many health facilities in sub-Saharan Africa, power is supplied from a combination of grid, generator, and solar sources. nLine's sensors take voltage and frequency measurements every 2 minutes but we don't know which power source is active at a given time.

This creates a challenge: which power source is currently supplying the load?

The ability to do so unlocks rich performance metrics — from solar PV and generator usage time (linked to fuel cost and carbon emissions) to evaluating the effectiveness of solar investments.

nLine sensors were deployed in different rooms in a number of hospitals in Sierra Leone and have been collecting data on voltage and frequency.

Your Challenge

> ***You'll take on the role of a data scientist at nLine tasked with improving the existing rule-based power-source classification algorithm. You will experiment (reasonably) with different Machine Learning models, and select which one works best for the power source detection problem. Using the selected model, you will label each two-minute time window with the most likely power source (solar, grid, generator). Finally, you will determine solar, generator, and grid usage durations for each of the rooms in the hospitals.***

Data Provided

You'll receive time series voltage and frequency data (two-minute resolution) from rooms in selected healthcare facilities in Sierra Leone along with the known power-source configurations supplying the room(e.g., “grid + solar”, “grid + gen”). This data will also contain the power source detected to be supplying the room at each time window, a detection done by the rule-based algorithm. Treat the detected power source column as the ground truth against which you will compare the results of your ML-based algorithm.

You will also receive a description of the current rule-based power-source detection method.

Guiding Questions

- Understand and review the rule-based power-source detection algorithm. What are its strengths and shortcomings?
- Visually explore the frequency and voltage data. Are there any distinct statistical or visual features that can be used to distinguish between power sources?
- Feature engineering: What derived features(from voltage and frequency) can be used to determine the power source supplying the load at any time?
- List different supervised Machine Learning models that can be applied to this power-source detection problem and explain why each would be appropriate.
- Train the ML models using the provided dataset and perform model evaluation.
- Report on which one performs best.
- What would a hybrid ML + rule-based version of this algorithm look like?
- Outlining next steps: how to improve accuracy, handle edge cases, etc.

Deliverables

- A Jupyter notebook with analysis, visualizations, model training, evaluation, selection.
- A detailed report responding to the guiding questions.
- A slide deck to present your work to nLine.

The Rule-Based Power-Source Detection Method

for Case study 3

1. Introduction

nLine's Gridwatch sensor reports data every two minutes. Each of these sensors reports data from health faclity connected to either solar, grid, generator, or some combination of these. The data reported is a timeseries dataset by each sensor indicates frequency and voltage, and whether power is on or off (and other metadata fields) for every two minutes. For the time slots in the timeseries where the power is off (i.e. outage), there is a ***null*** frequency value. The goal of the power source detection algorithm is to determine which power source is connected at any point in time which would effectively enable computation of solar, grid, and generator usage time in each site (i.e. health facility in SL). Understanding how much each power source gets used is important for economic planning of health facilities as well as monitoring the impacts of solarization projects. By understanding generator usage time, the fuel consumption can be computed hence the impact of solarization on the fuel consumption can be determined as an economic output of such development projects . We therefore to aim classify each sensor's data point as either ***solar***, ***grid***, or ***generator.***

2. Power Source Detection Algorithms

Previously, we have explored more automated approaches to detect the active power sources, including unsupervised ML algorithms and Kalman filtering. The ML model would be trained on the available sensor data features like frequency, voltage, and time, and augment them with derived variables through feature engineering. Additional inputs fed into the model could include time of day, known power source equipment at each location, and

more. By training on the unlabeled timeseries data, the ML algorithm clusters the data points to find inherent patterns that map to solar, grid or generator states.
While these methods would be generalizable, due to some complexities that exist in the data, these methods would perform less robustly by underfitting the data. To overcome these limitations, we developed a lightweight algorithm based primarily on expert knowledge and exploratory data analysis. In this post, we explain the methodology and some of the advantages of our approach.

3. Statistical Rule-Based Power Source Detection Algorithm

The core methodology focuses on analyzing frequency and feature-engineered variables from the sensor-reported frequency data. Based on exploratory data analysis, domain knowledge of the local grid, understanding of how different power source technologies work, and statistical methods, we are able to characterize the frequency and its features hence label the data points as solar, grid, or generator.
In the sections below, we describe the variables used for classification — frequency and other features derived from frequency — and explain the knowledge of local grid connection that help in the classification process.

3.1. Defining Variables for Classification

3.1.1. Frequency

This is the frequency value as reported by the sensor. The expected nominal frequency for a Sierra Leone grid is 50Hz. From our exporatory data analysis of the data collected so far from SL as well as knowledge of the working of inverter-based systems:

- Solar systems exhibit frequency with a tight distribution around 50Hz.
- Grid frequencies hover near 50Hz but have more variance.
- Generator frequencies vary widely, have a wider deviation from 50Hz nomina, with some staying as low as ~43Hz and some as high as ~57Hz.

With this in mind, we create frequency thresholding rules to distinguish between the power sources.

3.1.2. Frequency deviation from the electrically closest grid-only-connected facility

Frequency alone is not always enough, as the ranges can overlap at times, rendering the thresholding rule alone inadequate. To tighten up the logic around grid classification for locations where grid is one of the power sources, we use the frequency deviation from the electrically nearest grid-only connected facility. We determine the electrical proximity by eyeballing frequency profiles and analyzing the frequency patterns. Electrically proximal locations usually have the same frequency profile.
We compare the sensor-reported frequency to that of the current grid-frequency. The closer the frequency is to the current grid frequency, the more likely it is that it is a grid frequency (For a synchronized grid, frequency measured anywhere on the grid at any point in time should be the same value). We set a maximum deviation threshold of 0.11Hz between the sensor-reported frequency and the current grid frequency to classify the frequency as “grid”. We determine the current grid frequency by computing the median frequency from sensors in rooms that are grid-only connected, and connected to the same grid.

3.1.3. Frequency spread/variability of the sensor data within a time window

We use the mean absolute deviation (MAD) of the sensor frequency timeseries within a sliding time window. The time window used is experimental based on the exploratory data analysis we’ve conducted on the SL data and is currently ~15 minutes (3 data points before and 3 data points after the current data point). Solar has the lowest variability while generators have the highest variability. We use a threshold of 0.1 neighborhood MAD for solar.

3.1.4. Label of the combination of power sources in the room

3.2. Other Considerations for Classification - Ground Truth Knowledge

For the case of SL, there is some ground-truth knowledge that helps in both the classification steps and the fine-tuning steps to ensure correctness. The

qualitative data collected during deployment labels each room where a sensor is deployed based on the power source it is connected to.

- ***Knowledge of grid-only connected rooms in health facilities*** - as described above, we used grid-only connected rooms in the facilities as grid-references. Some of the rooms are connected to grid only hence we know any data coming from these are from the grid and are used as grid reference.
- ***Knowledge of generator-only or solar-only connected rooms -*** While the use of the variables/features described above help in classification, some of the rooms are only connected to one power source so they are automatically labelled as such.

The use of variables defined above for classification is therefore most useful for rooms where there are multiple power sources in use.

3.3. Classification Steps

1. Label as outage if*:*
    a. the power is out
2. Label data point as grid if**:**
    a. not outage
    b. the power source is grid only **OR**
    c. one of the power sources is grid and the frequency deviation from the geographically closest grid median is ≤ 0.11
3. Label data point as solar if**:**
    a. not outage or grid
    b. frequency value is between 49.8 to 50.2
    c. MAD within the time window is ≤ 0.1
    d. not one of the rooms we know solar isn't installed.
4. Label all others as generator
5. Perform smoothing

3.4. Smoothing

Some smoothing steps are applied based on some combination of: post classification exploratory analysis, knowledge of the working of solar/grid/generator, ground truth knowledge based on survey data as well as interaction with health facility (HF) officials through our project manager, contextual understanding of the grid network in SL, and observation of historical data in SL.

1. At PCMH, only allow for generator label when grid is off.
2. For Ola During, when the grid is off, set to solar if the frequency is within solar threshold.

We continue to incorporate some smoothing steps (not included here) as we keep learning about the data, SL context, HF-specific knowledge, and making intuitive observations.

4. Limitations

While the methodology leverages domain knowledge, understanding of the grid context, exploratory analysis, and feature engineering to correctly classify power sources, it would be beneficial to be able to device a generalizable algorithm that can be applied in different contexts. The main limitation is that the method is specific to current Sierra Leone hospitals where nLine sensor deployments are, making the method less transferable without adjustment. It is therefore not generalizable to other contexts.

5. Possible Improvements

The current power source detection algorithm is effective to a large extent, but suffers the limitations stated previously. There are ways we can continue improving the methodology to make it even more effective:

- We can use this rule-based approach to reliably label new data over longer time periods. This ground truth historical dataset can then be used to train more advanced ML models in a supervised manner, rather than relying purely on clustering unlabeled data.

- Feature engineering steps used in our methodology also provide a solid starting point for predictive ML algorithms. Variables like frequency distribution statistics, variability, and geospatial correlations/grid distances would be useful features for a supervised ML method.
- We can run these new ML approaches in parallel to our current system and continuously check performance against our labeling to ensure consistency and correctness. We would therefore only rely on ML if the accuracy of the ML model is acceptable.